Tripple 7 Movie Empire Review

By Group 7

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Business Understanding

Business Problem

- Tripple 7 wants to enter the movie industry
- It lacks knowledge of what makes movies successful
- This is a high risk of financial loss without data insights

Why this matters:

- Poor investment decisions could lead to major losses

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Project Objectives and Questions

Objectives

- To determine the movie genres that are the most profitable.
- To determine the best time to release a movie for maximum returns.
- To determine whether higher-rated movies generate more revenue.

Questions

- Which movie genres are the most profitable?
- When is the best time to release a movie for maximum returns?
- Do higher-rated movies generate more revenue?.

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Data Sources

- We agreed to use two key datasets as they were the most relevant to answering our research questions:

1. TMDb dataset (tmdb.movies.csv)

as it Provides movie details (genres, ratings, popularity)

1. The Numbers dataset (tn.movie_budgets.csv)

as it Provides financial data (budget, revenue)

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Data Understanding

- 2316 movies
- Key variables:
- Genre
- Budget
- Domestic & worldwide revenue
- Ratings

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Data Understanding

What we did:

- Merged TMDb + The Numbers datasets
- **Joined using:**

original_title (TMDb) and movie (The Numbers)

- **Result:**
- Final dataset: 2316 movies

Why we did this:

- TMDb gives ratings & genres
- The Numbers gives financial performance
- Combining them allows us to answer our research questions.

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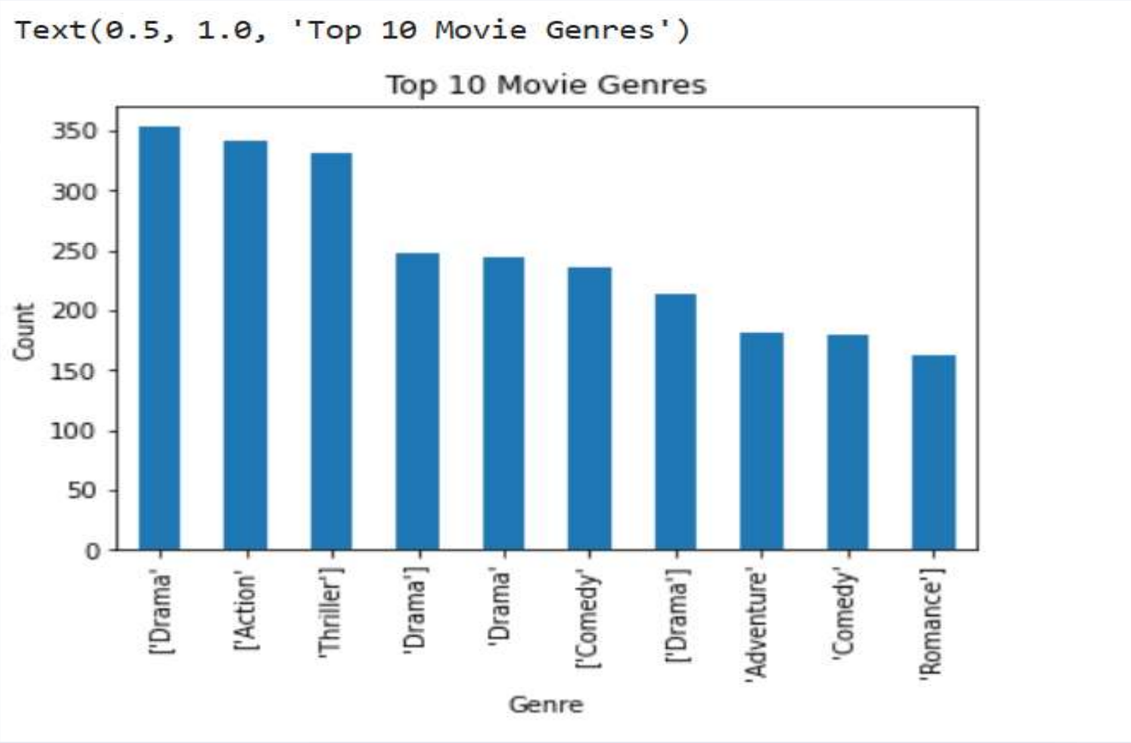
Data Understanding

Top 10 Movie Genres

After analysis of the top 10 movie genres, the top most movie genre was drama, and the last was Romance.



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Data Preparation

Data Cleaning & Preparation

- Converted revenue columns from text to numeric
- Checked for missing values, where none was found
- Removed duplicates

Feature Engineering

- **Created**:
- total_profit = worldwide_gross - production_budget
- release_year from date
- Genre mapping (IDs → names)

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Analysis

Research Question 1

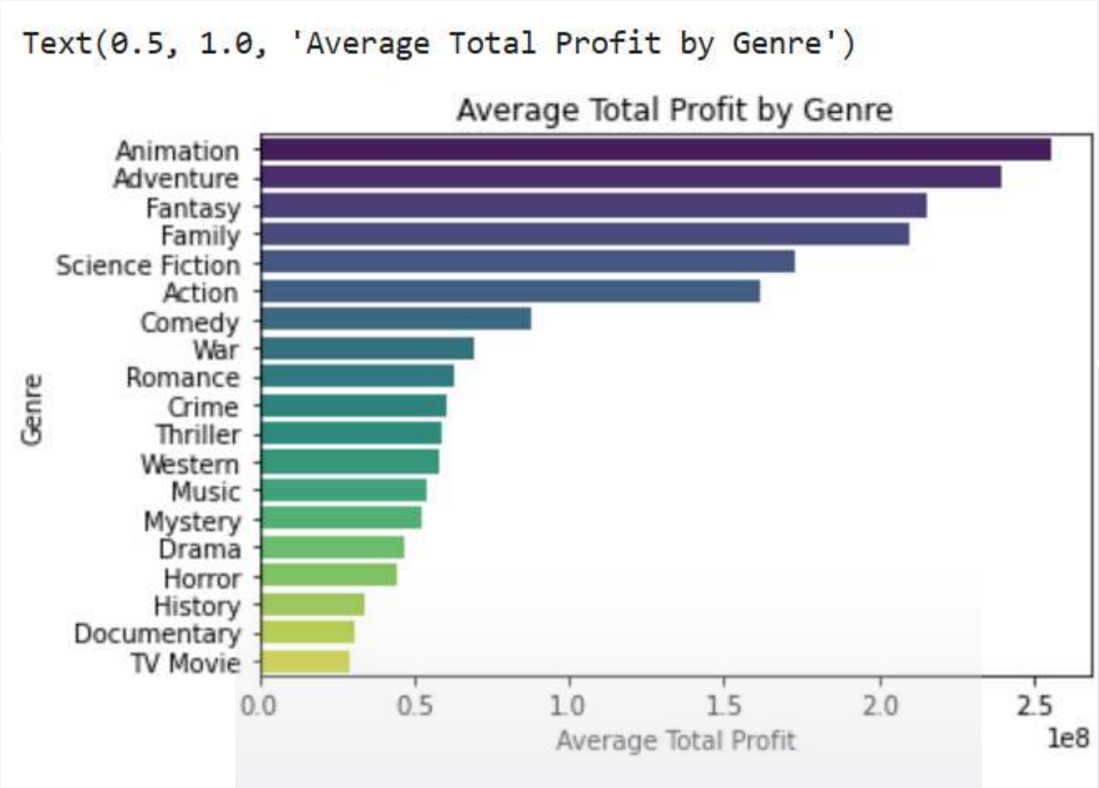
Which movie genres are most profitable?

Top genres:

- Animation
- Adventure
- Fantasy

Business Insight:

- Genre strongly influences profitability
- High-performing genres consistently generate higher returns



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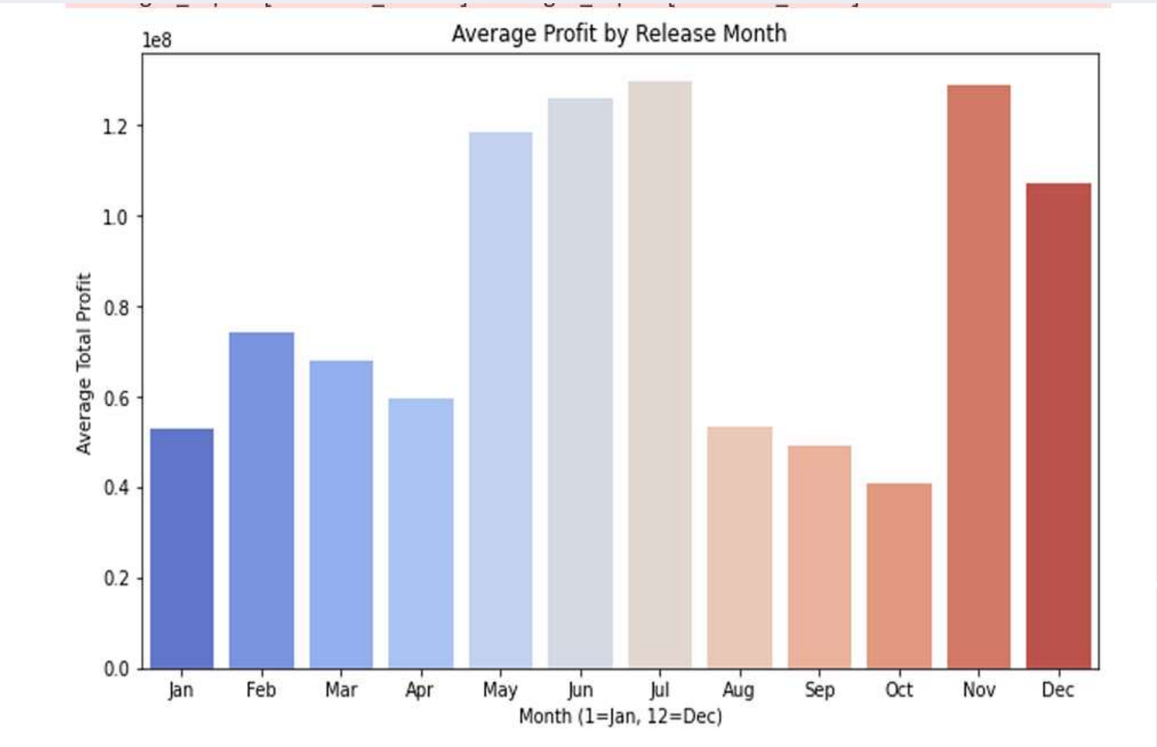
Analysis

Research Question 2

When is the best time to release movies?

•Movies released in summer (May–July) generate the highest profits
•Secondary peak occurs in holiday season (Nov–Dec)
•These periods align with:
•School holidays
•Increased leisure time
•Higher audience turnout



Business Insight:
Releasing movies during peak seasons maximizes box office revenue and audience reach

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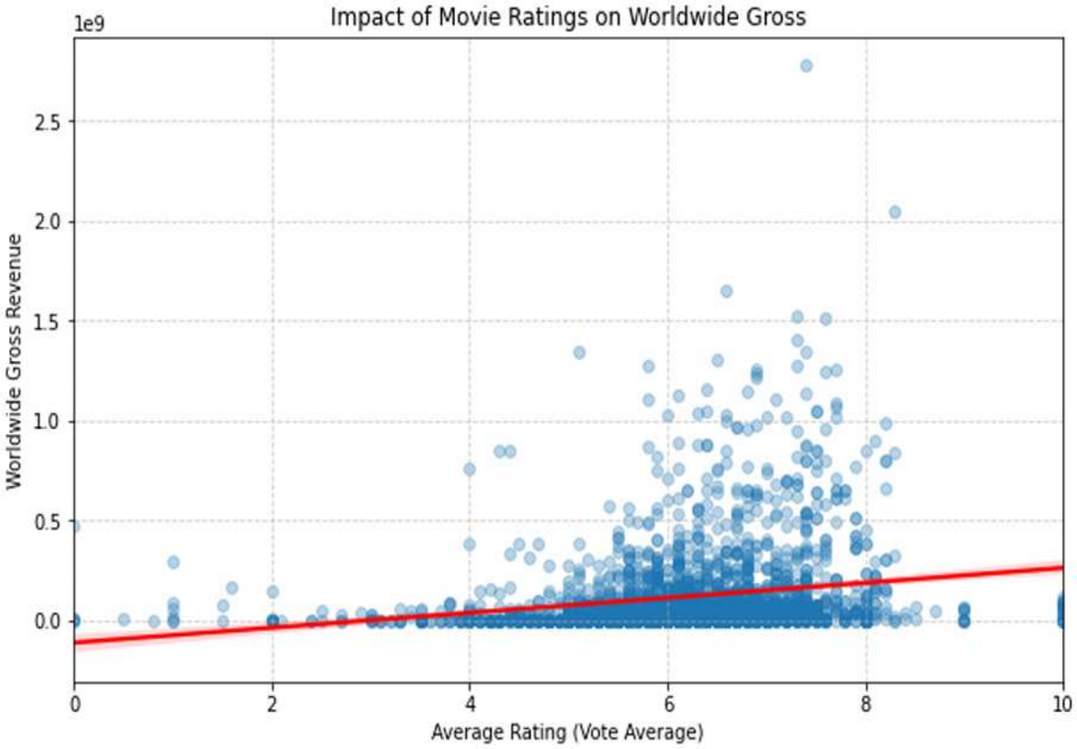
Analysis

Research Question 3

Do higher-rated movies generate more revenue?

•**RESULTS:**
•Weak correlation between ratings and total profit (0.20)
•Highly rated movies do NOT always generate high revenue
•Other factors (genre, release timing, marketing) have stronger impact



Business Insight:
High ratings alone do not guarantee success.
Hence, focus on strategy, not just quality

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key Insights

- Genre is the strongest driver of profitability
- Animation, Adventure, Fantasy dominate revenue
- Release timing significantly impacts success
- Ratings have minimal influence on revenue
- Industry growth driven by technology and streaming trends

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Recommendations

1. **Invest in high-performing**
- genres: Animation, Adventure, Fantasy,

2. **Schedule releases during:**
- Summer (May–July), Holiday season (Nov–Dec)

3. **Focus on:**
- Strong marketing campaigns, Audience targeting strategies, High production quality.

4. **Avoid relying solely on ratings as a success metric**

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Conclusion

- Data analysis provides clear guidance for Tripple 7 to enter the movie industry.
- Success depends on:

Choosing the right genre, Timing releases strategically, Understanding audience behavior

- By applying these insights, Tripple 7 can:

Reduce financial risk, Increase profitability, Make data-driven decisions.